

Numerical Programming

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AP#6

Compute volume from digital image

Problem 6.1

- ▶ Select digital image of an object with axial symmetry.
- ▶ Post selected problem in the chat
- ▶ Allowed: built-in libraries
- ▶ Requirement: detect edges, approximate with a parametric suitable part, consider virtual object reconstruction using 360 degree rotation of the curve around the axe of symmetry; compute volume, visualize.
- ▶ Deliverables:
 - ▶ working code
 - ▶ 2 page paper explaining your work: model, approach, experiment, conclusions
 - ▶ 3 min video for TA with explanation and demonstration of your work
- ▶ Deadline: 5 days after publication of AP

Important Notice

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- ▶ Test your code, it should work properly for input data assigned by TA or instructor.
- ▶ AP assigned 0 points if:
 - ▶ a model problem provided twice by students. Make sure, your model is different from models provided by others.
 - ▶ submitted results are not reproducible
 - ▶ student cannot apply his own code for data provided by TA or instructor